

**SECP3623 – DATABASE PROGRAMMING**  
**SEM I – 2025/2026**  
**MongoDB(shopdb.orders)**

**Instruction:**

Insert the following data into the shopdb database and the orders collection.

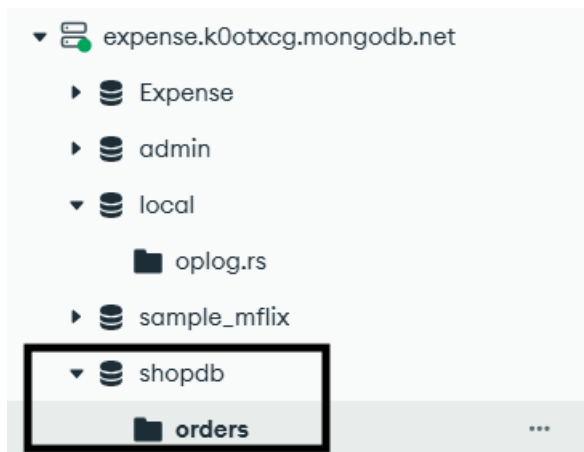
use shopdb

```
db.orders.insertMany([
  {
    order_id: "ORD101",
    customer: "Ali Ahmad",
    product: "Laptop",
    quantity: 1,
    price: 3500,
    status: "Completed",
    order_date: new Date("2025-01-05")
  },
  {
    order_id: "ORD102",
    customer: "Siti Aminah",
    product: "Smartphone",
    quantity: 2,
    price: 2800,
    status: "Pending",
    order_date: new Date("2025-01-06")
  },
  {
    order_id: "ORD103",
    customer: "John Lee",
    product: "Tablet",
    quantity: 1,
    price: 1800,
    status: "Completed",
    order_date: new Date("2025-01-07")
  },
  {
    order_id: "ORD104",
    customer: "Nur Aisyah",
    product: "Laptop",
    quantity: 1,
    price: 3700,
    status: "Cancelled",
    order_date: new Date("2025-01-08")
  },
  {
    order_id: "ORD105",
    customer: "Ahmad Zaki",
```

```

    product: "Headphones",
    quantity: 3,
    price: 450,
    status: "Completed",
    order_date: new Date("2025-01-09")
  },
  {
    order_id: "ORD106",
    customer: "Farah Izzati",
    product: "Smartwatch",
    quantity: 2,
    price: 1200,
    status: "Pending",
    order_date: new Date("2025-01-10")
  }
])

```



a Write a MongoDB command to display all orders with status "Completed".

```
> db.orders.find({ status: "Completed"})
```

```

< {
  _id: ObjectId('695f13e339cf958c4fc3e382'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e384'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e386'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
}

```

- b) Write a MongoDB command to display all orders with price greater than RM2000.

```

> db.orders.find({ price: { $gt: 2000 } })

```

```

< {
  _id: ObjectId('695f13e339cf958c4fc3e382'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e383'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e385'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}

```

- c) Write a MongoDB command to display all orders sorted by price in descending order.

```

> db.orders.find().sort({price: -1})

```

```
< {
  _id: ObjectId('695f13e339cf958c4fc3e385'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e382'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e383'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
```

```
{
  _id: ObjectId('695f13e339cf958c4fc3e384'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e387'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e386'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
}
```

- d Write a MongoDB command to display all "Pending" orders sorted by order\_date  
) (latest first).

```
> db.orders.find({status: "Pending"}).sort({order_date: -1})
```

```
< {
  _id: ObjectId('695f13e339cf958c4fc3e387'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695f13e339cf958c4fc3e383'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
```

- e) Using aggregation, calculate the total sales amount (price) for each product.  
)

```
> db.orders.aggregate([
  {
    $group: {
      _id: "$product",
      totalSalesAmount: { $sum: "$price" }
    }
  }
])
```

```
< {
  _id: 'Headphones',
  totalSalesAmount: 450
}
{
  _id: 'Tablet',
  totalSalesAmount: 1800
}
{
  _id: 'Smartphone',
  totalSalesAmount: 2800
}
{
  _id: 'Smartwatch',
  totalSalesAmount: 1200
}
{
  _id: 'Laptop',
  totalSalesAmount: 7200
}
```

- f) Using aggregation, calculate the total sales amount for orders with status "Completed".

```
> db.orders.aggregate([
  { $match: { status: "Completed" } },
  { $group: { _id: null, totalSalesAmount: { $sum: { $multiply: ["$price", "$quantity"] } } }
])
< {
  _id: null,
  totalSalesAmount: 6650
}
```

- g) Write a MongoDB command to update the status of order ORD102 from "Pending" to "Completed".

```
> db.orders.updateOne(
  { order_id : "ORD102" },
  { $set: {status: "Completed"}}
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- h) Write a MongoDB command to increase the quantity of order ORD105 by 1.  
)

```
> db.orders.updateOne(
  { order_id: "ORD105" },
  { $inc: { quantity: 1 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- i) Write a MongoDB command to delete the order with order\_id "ORD104".

```
> db.orders.deleteOne({order_id: "ORD104"})
< {
  acknowledged: true,
  deletedCount: 1
}
```

- j) Write a MongoDB command to delete all orders with status "Cancelled".

```
> db.orders.deleteMany({status: "Cancelled"})
< {
  acknowledged: true,
  deletedCount: 0
}
```